

➤ **Innate immunity VS Adaptive immunity**

Innate immunity :

The innate immune system is essentially made up of **barriers** that aim to keep viruses, bacteria, parasites, and other foreign particles out of your body or limit their ability to spread and move throughout the body.

Adaptive immunity:

(**acquired**) immunity a specific defense mechanisms effective against specific microorganism.

It depends upon having previous contact with a particular microorganism (memory). It is mediated either through antibodies (humoral) or cytotoxic effects (cellular)

- T-lymph: intercellular
- B-lymphocytes: extracellular

Types :

- **Active**
- **Passive**

➤ Types of immunity:

	Active immunity	Passive immunity
Produced by	Individual own immune system	An animal or human antibody and transferred
Acquired by	1- Active disease = Antigen(Ag) 2- Vaccination (Live, attenuated – Inactivated)	1- Antibody (Ab) = Immunoglobulins (Ig)
Protection & onset	Protection through memory varies ..slow onset.. long lasting	Immediate & temporary protection

➤ Types of vaccines (active immunity):

	Live ,attenuated vaccine	Inactivated ,killed vaccine
Produced by	Modifying the virus or bacteria to give immunity	Fraction of the virus or the bacteria such as cell wall polysaccharides attached to protein carrier
Disease	Do not produce disease (but may)	Unable to produce disease
Notes	Not given in immunocompromised & pregnancy	Can be given in immunocompromised
	↓ virulence of pathogen by attenuation by heat or light	

➤ Live attenuated vaccines **ROME Is MY Best Place To go Yet**

ROME	• Rubella -Rota • Oral Polio & Oral Typhoid • Measles & Mumps • Epidemic typhus
Is	• Influenza (LAIV)
Best	• B CG
Place	• P lague
Y	• Y ellow fever
Varicella,, zoster	

Live vaccine contra-indication

- *Don't give live vaccines to pt who have CD4 < 200*
 - Not given in pregnancy
 - Not given in immunocompromised (chemotherapy,,HCT,,biologic drug)
- a.Patient receiving cancer chemotherapy (**3 months** separation)
b.Recent hematopoietic cell transplant (HCT)
c.Biologic agents (TNF inhibitors, Rituximab, Etanercept d- All vaccines with CD4 < 200 cells/mm³



➤ Live vaccine with corticosteroids:

Topical steroids	Any dose	• Live vaccines given immediately
Systemic steroids	less than 2 mg/kg/day "low or moderate dose"	• Live vaccines given immediately
	≥ 2 mg/kg/day for less ...than 14 days	• Live vaccines given immediately
	≥ 2 mg/kg/day for more than 14 days	• Live vaccines should be delayed at least 1 month after stopping the steroid.



InFLUENZA(FLU)

<i>Inactivated influenza vaccine (IIV)</i>	<i>Live attenuated influenza vaccine (LAIV)</i>
Can be given with Influenza A ntiviral D rug (IAD) like Tamiflu (Oseltamivir)	Not given with Influenza A ntiviral D rug (IAD) like Tamiflu (Oseltamivir)
	- IAD cessation → wait 48 hours → Take LAIV - LAIV → wait 2 weeks → Take IAD - If IAD administered within 2 weeks of LAIV, the vaccine dose should be repeated 48 hours after the last dose of antiviral medication

⊗ LAIV should not be given to:

- 1- Patient with severe allergic reaction to LAIV
- 2- Patient allergic to eggs
- 3- Pregnant woman
- 4- Child (2:17) receiving aspirin or aspirin containing
- 5- Child (2:4) with asthma or wheezing in past year
- 6- Immunosuppressed patient
- 7- Patient taken antiviral in the previous 48 hours

- Influenza viruses undergo **shift and drifts**; therefore, every year new influenza (flu) vaccine is formulated and must be **administered every year**.
- New influenza season start approximately in **November (11)** every year
- The new vaccine must be **started in October (10)** as it takes about 2 weeks after vaccination for antibodies to develop and provide protection.
- ★ **Child less than 9 years** given **2 doses** in one flu season, one is a booster dose. → (In 1st time only)
- ★ Influenza vaccine normally taken once per year but it can be taken twice (every 6 months) in high risk patients.
- ★ High risk patients include (COPD, HTN, Dyslipidemia, Female, Old)
- ⊗ Not used for **children (0:6) months**
- ★ Influenza vaccine is given every year even if you do not need it.
- ★ Inhaled influenza vaccine → **FluMis**

➤ **Role of Passive immunity:**

- 1- Prophylaxis of infectious disease (tetanus ,hepatitis)
- 2- Prophylaxis therapy (after varicella exposure)
- 3- Treatment of Antibody deficiency (leukemia pt)

1-Prophylaxis of infectious disease (tetanus ,HIB)

- Tetanus immunoglobulin(TIG)

	Unknown history	Known history
Minor wounds (10)yrs	Td	• If last Td dose received within 10 years → <u>Notreatment</u> • If last Td dose received more than 10 years → Td only
Major wounds (5)yrs (deep) (contaminated)	Td + TIG	• If last Td dose received within 5 years → <u>Notreatment</u> • If last Td dose received more than 5 years → Td

- Hepatitis B immunoglobulin(HBIG)
 - 1- Pregnant woman with Hepatitis B when she delivers the baby must be given → Hepatitis B vaccine (HBV) + HBIG
 - 2- Nurse infected with Hepatitis B patient blood must be given → Hepatitis B vaccine (HBV) +HBIG
 - 2- HBV + HBIG → administered at different body sites.

2-Prophylaxis therapy

- Varicella zoster immunoglobulin (VZIG) → Given to patients (Leukemia or ↓ immunity) exposed to varicella zoster

3.Treatment of Antibody deficiency

- 1st Immunodeficiency or as chronic lymphocytic leukemia must be given →
- **IVIG**(intravenous immunoglobulin) or **IGIM**(Intramuscular immunoglobulin) every 2 to 4 weeks to maintain immunity

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- 2 inactive vaccines: can be taken at same time
- Live and inactive: same time
- 2 Live vaccines: either same time or 28 days apart

➤ Vaccine storage:

- Most vaccines require(bottom shelf) refrigeration ($2^{\circ} - 8^{\circ}\text{C}$) except MMR can be stored in freezer.
- **ZOSTRAVAX** required to be frozen ($\leq -15^{\circ}\text{C}$) **OPV** $\gg -20^{\circ}\text{C}$

Vaccine	Stability	Temperature	Comment
Tetanus (T)	1 hour.	2 – 8	
MCV	3 hours	2 – 8	Protect from light
OPV	3 hours	-20	
BCG	4 hours	2 – 8	Protect from light

- Protect from light → MCV, BCG, Measles
- OPV after opening can be kept (2-8) for 6 months

➤ Vaccines administration routes:

➡ Majority are **IM** only except

IM or SC	SC only	Oral only
• IPV → Inactivated poliovaccine • PPSV → Pneumococcal Polysaccharide Vaccine	• MMR • Varicella • ZOSTRAVAX	• OPV → Oral poliovaccine • RV → Rota virus vaccine

➤ **Vaccination and pregnancy:** → (Tdap ,IIV) HBV

Vaccination prevents pregnant: Depo-Provera (DMPA)

Vaccines given in pregnancy	Vaccines contraindicated in pregnancy
1- Tdap: During each pregnancy ideally between 27, 36 weeks gestation (Last 3months) 2- IIV (Inactivated Influenza Vaccine) → give immunity to child during pregnancy and 6 months after birth.	1- HPV 2- MMR 3- Varicella 4- ZVL 5- BCG 6- LAIV ➤ All live vaccines

Tdap → pregnancy ,adult.
 Dtap → <7 years children.
 DT → < 7 who had allergy from pertussis.
 Booster dose every 10 years.

★ Pregnant woman is more liable for influenza complications.

➤ **Vaccination and Antibiotics:**

- Some vaccines may contain small amount of certain antibiotics (neomycin, streptomycin, gentamycin, polymyxin B) to prevent bacterial contamination during manufacture process.
- MMR, IPV,ZVL**
- ☒ Patients with allergy to these antibiotics **must not** be given these vaccines.

➤ **Vaccination and egg allergy:** (,,influenza ,, yellow fever ,MMR??)

Latex: HPV, RV

Gelatin: varicella, MMR, Yellow fever, Zoster

➤ Vaccinations before Haj:

- 1- Annual influenza **flu**
- 2- vaccination **MCV**
(Meningococcal)

+Polio+ Corona + Zika

Emergency staff: should take meningitis vaccine

➤ Saudi MOH vaccination schedule:

Birth	 B (Hep B) → 12 hours from birth	• A = HepA • B= HepB • D =DTaP • H =Hib • I =IPV • O =OPV • M =MCV4 • Mea =Measles • MMR =MMR • P = PCV • R =RV • V =Varicella
2 months	 DR BHIP	
4 months	 DR BHIP	
6 months	 DO BHIP BCG	
9 months	 MeaM	
12 months	 MMR MOP	
18 months	 MMR DOV AH	
24 months	 A	
School entry (4-6) years	 MMR DOV	

➤ **Adverse effects of vaccines:**

1- Injection site reactions:

(Pain, swelling, Erythema "redness") → **Major**

2- side effects:

Fever

- It is important to monitor patients for at least **15 minutes** following administration.
- Severe allergy handled with **Epinephrine, diphenhydramine** and **CPR**

➤ **Key considerations in special population:**

★ **Preterm birth**

- Immunize based on chronological age
- Do not reduce the recommended dose
- Delay Hep. B vaccine until the baby is:

➔ More than **2 kg** body weight

➔ More than **30 days** old

Less than 7 years „children“	✓ D Tap
Above 7 years & Pregnancy	✓ TD –Tdap
Adult	✓ Adult should receive one Td booster every 10 years. ✓ Adult should replace 1 Td booster with a Tdap dose once in their lifetime.

- Td vaccine wears off after 10 years

Hepatitis

Hep. A	- Hep A vaccine is recommended for travel to most parts of the world. - Total doses =2
Hep. B	- Hepatitis B vaccine is given in a 3-doses series at 0, 1, 6 months. - Minimum duration 4 weeks between two vaccines. - If a 2nd dose vaccine was <u>given too soon (before minimal interval)</u> → <u>don't count</u> this incorrect dose and <u>repeat</u> it after the minimal time has passed since the incorrect dose. - Total doses =3 - Pregnant woman with Hepatitis B when she delivers the baby must be given → (HBV) +HBIG - Nurse infected with Hepatitis B patient blood must be given → (HBV) +HBIG - HBV + HBIG → administered at different body sites. <u>Preterm birth</u> → Delay Hep. B vaccine until the baby is: 1- More than 2 kg body weight 2- More than 30 days old
Hep. C	There is no vaccine available to provide protection against hepatitis C

Varicella zoster

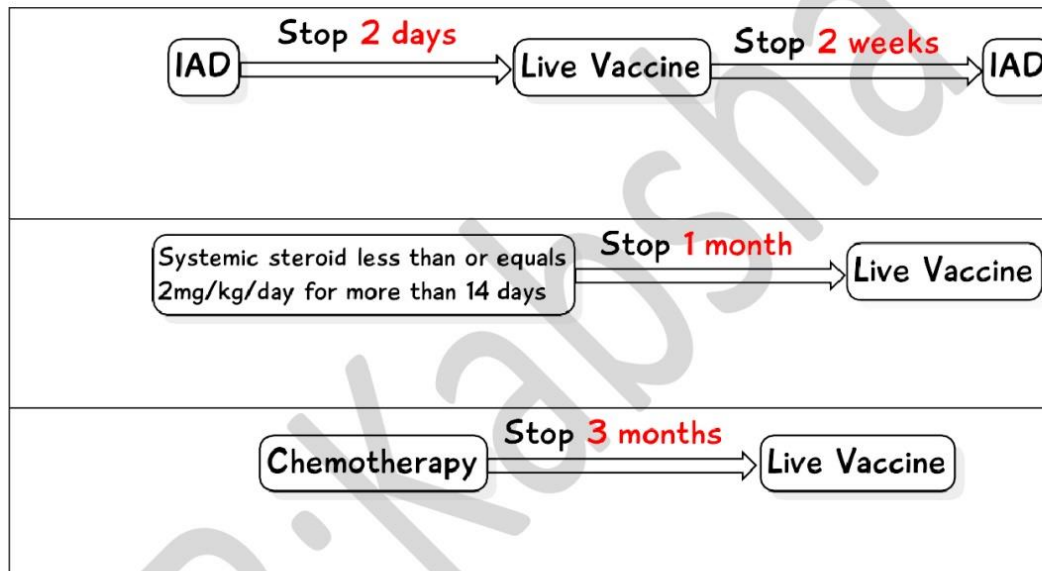
- Zoster vaccine **must not** be given to the patient **currently treated** for **shingles** or post-herpetic neuralgia. It can be administered once the **symptoms are resolved**. (Disease symptoms are cleared)
- Varicella zoster immunoglobulin (**VZIG**) → Given to patients (Leukemia or ↓ immunity) exposed to varicella zoster

Others

HPV	❖ <u>Human Papilloma virus vaccine</u> 1- ↓ incidence of <u>infertility</u> 2- ↓ incidence of <u>cervical cancer</u> 3- can cause <u>syncope</u> → patients remain seated for at least 15 minutes after receiving the vaccine. 🌀 Brand name → Cervarix (bivalent), Gardasil (quadrivalent)
RV	1- The 1st dose of RV → before 14 weeks 2- The last dose of RV → before 8 months
Hib	☒ Haemophilus influenza vaccine should not be given over 5 years child.
Hg	☒ Thimerosal (Mercury) containing vaccine → Autism
BCG	✓ BCG vaccine has the maximum age ??? → Life-long immunity
PCV - PPSV	✓ Pneumococcal vaccine given to patients older than 65 years
MCV	✓ Emergency staff at hospital should receive meningitis vaccine.

Notes

- ★ Taking antibiotics shouldn't prevent vaccination.
- ★ Antibiotics will not affect how your child's body responds to vaccine.
- ★ Cyclosporin is an immunosuppressant given after organ transplant.



DM:

DM pt.: Pneumococcal, HBV, Influenza

Diabetic foot: TD only

Emergency staff: should take meningitis vaccine

Hep A is the most recommended for travel

Older pt:

-> 50: shingle

-All pts.> 65: PCV13 then after 1 year give PPSV23 then 5 years from last dose

LYMERix vaccine use to prevent Lyme disease

Lyme disease caused by bacterium *Borrelia*

Airborne virus need vaccine or cause Endemic: MMR

<u>Disease</u>	<u>Cause /pathogen</u>
Diphtheria	Corynebacterium
Whooping cough	Pertussis vaccine DTap - TDap

Lyme	by tick by bacteria borreliaburg
Plague fever	By Flear by <i>Yersinia pestis</i>
Dengue fever	mosquitos
Anthrax	Cattle
Dental plaque/carries	<i>Streptococcus mutans</i>

Varicella (chickenpox)varicella vaccine

Herpes zoster (shingle)shingrix

Prevention of cervical cancer and reduce the incidence of infertility: HPV (Gardasil

)it causes syncope à let the pt. set for 15 min.

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0020106401621